

Saccharometer DI Water Glucose Solution Yeast Suspension

2	24 ml	0 ml	4 ml
3	12 ml	12 ml	4 ml
4	4 ml	12 ml	12 ml

Employing Steps in the Scientific Method:

1. Record the **Question** that is being investigated in this experiment.

2. Record a **Hypothesis** for the question stated above.

3. Predict the results of the experiment based on your hypothesis (if/then).

4. Perform the experiment below and collect your data.

Procedure:

1. Prepare yeast suspension: Add 7 grams yeast to 50 ml warm tap water. Stir to mix. Alternatively, you can use the yeast suspension from Part 2. Optional: Add a few drops of red food coloring to the yeast to increase contrast, allowing easier measuring of the height of yeast in saccharometers.
2. Label 4 test tubes and 4 saccharometers # 1- 4. Use a transfer pipette to add the appropriate amount of glucose and distilled water listed in Table 2 to the corresponding labeled test tubes.
3. Use a transfer pipette to add the appropriate amount of yeast solution listed in Table 1 to the corresponding labeled test tubes. It is important to work carefully and quickly after adding the yeast solution to the glucose and water.
4. Carefully pour the contents of the test tubes into the correspondingly labeled saccharometer, ensuring that the solutions are well mixed.
5. Carefully tilt the saccharometers to allow any air bubbles that are trapped in the arms of the vertical tube to escape.
6. Begin the timer for the experiment and measure the size of any bubbles (in mm) that are trapped in the vertical arms of the saccharometers. Record this measurement as the 0 time point.
7. Position the saccharometers on the large plastic tray, positioning them around a plastic weigh boat to catch any fermentation overflow that may occur.